

CE 310: Strength of Materials and Structures

Course Code	CE 310
Course Name	Strength of Materials and Structures
L-T-P-C	3-0-0-3
Prerequisites	IC240: Mechanics of Rigid Bodies
Intended for	UG
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Module 1:** Introduction to loads, structural materials, and support conditions, Overview of internal and external stability, Definition of determinate structures. (3 Lectures)
- **Module 2:** Mechanics of small deformation: Free Body Diagram, Concepts of stress and strain - types and definition, stress-strain characteristics of ductile and brittle materials, elastic constants and their relationships, Axially loaded members, Concept of Torsion and Torsional behaviour of circular shafts. (8 Lectures)
- **Module 3:** Transformation of stresses and strains, principal stresses and strains, Mohr's circle, and Failure theories. (5 Lectures)
- **Module 4:** Determination of external and internal forces in Trusses, Cables, Arches, and Beams, Concept of shear force, and bending moment diagrams of determinate beams and frames. (10 Lectures)
- **Module 5:** Flexural and shear stresses in beams, deflections of beams using double integration, introduction to moment area, unit load, and conjugate beam methods. (8 Lectures)
- **Module 6:** Influence lines for statically determinate structures; Moving loads on beams and trusses; Maximum shear force and bending moment due to moving loads. (4 Lectures)
- **Module 7:** Columns: Euler's theory, Critical load for different end conditions, eccentric loading, columns with small initial curvatures. (4 Lectures)

Text Books

1. Hibbeler, RC, **Structural Analysis**, 9th Edition, Pearson Education, 2017.
2. Hibbeler, RC, **Mechanics of Materials**, Pearson Education, 10th Edition, 2016.

References

1. Timoshenko, S.P., and Young, D.H., **Elements of Strength of Materials**, 5th Edition, 2003.
2. Beer, F.P., Johnston, E.R., DeWalt, J., and Mazurek, D.F.,
3. Egor P. Popov, **Engineering Mechanics of Solids**, 2nd Edition Pearson Education Inc, 2009
4. Gere and Timoshenko, **Mechanics of Materials**, 2nd Edition, CBS Publishers, 2004.
5. Reddy, CS, **Basic Structural Analysis**, Tata McGraw Hill, 2001.
6. Menon, D., **Structural Analysis**, Narosa Publishing House, 2008.
7. Ross, C.T.F, Case, J, and Chivler, L, **Strength of Materials and Structures**, 4th edition, Butterworth Heinemann, 1999.
8. Megson, THG, **Structural and Stress Analysis**, Butterworth-Heinemann, 2014.